Linear Stability To Check

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Step 0: The model

$$\partial_t N = N\,(c(1-l)\,I + dR - m) + D_N \nabla^2 N$$

$$\partial_t I = K - r\,I - c\,I\,N + D_I \nabla^2 I$$

$$\partial_t R = c\,l\,I\,N - r\,R - dRN + D_R \nabla^2 R$$

The story: cells consume I at rate cIN. A fraction l of that flux is leaked as the byproduct R; the remaining fraction $(1-l)$ becomes growth. The leaked byproduct is then re-consumed by the same population at rate dRN, also contributing to growth.

Step 1: Homogeneous steady state

Set all time derivatives and all Laplacians to zero.

From the I equation:

$$K = rI^* + cI^*N^* \quad \Longrightarrow \quad I^* = \frac{K}{r + cN^*}$$

From the R equation:

$$cl\,I^*N^* = rR^* + dR^*N^* \quad \Longrightarrow \quad R^* = \frac{cl\,I^*N^*}{r + dN^*}$$

These two denominators will appear everywhere, so name them now:

$$\boxed{u \equiv r + cN^*, \qquad v \equiv r + dN^*}$$

Each is a total removal rate: u is the rate at which I leaves the system (dilution r plus consumption cN^*), and v likewise for R. In these terms

$$I^* = \frac{K}{u}, \qquad R^* = \frac{clKN^*}{uv}$$

From the N equation, the bracket must vanish (taking $N^* \neq 0$):

$$c(1-l)\, I^* + dR^* = m$$

This is the equation that fixes N^*. We won't need to solve it explicitly — but we *will* use the fact that it holds, repeatedly.

Step 2: Perturb

Write

$$N = N^* + \nu\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}, \qquad I = I^* + \rho\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}, \qquad R = R^* + \sigma\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}$$

with ν, ρ, σ infinitesimal. Since $\nabla^2 e^{i\mathbf{k}\cdot\mathbf{x}} = -k^2 e^{i\mathbf{k}\cdot\mathbf{x}}$, every Laplacian becomes multiplication by $-k^2$.

Quasi-steady-state assumption: the resources equilibrate fast, so we set $\partial_t \rho = \partial_t \sigma = 0$ while keeping their diffusion. This is the key simplification: it lets us solve for ρ and σ in terms of ν, and reduces a 3×3 eigenvalue problem to a single scalar equation.

Step 3: Solve for ρ (the external resource response)

Linearise the I equation. The only nonlinear term is $-cIN$, which varies as $-c(I^*\nu + N^*\rho)$:

$$0 = -r\rho - c\,(I^*\nu + N^*\rho) - D_I k^2 \rho$$

Collect the ρ terms:

$$0 = -\underbrace{(r + cN^*)}_{=u}\,\rho - D_I k^2 \rho - cI^*\nu$$

$$\boxed{\rho = -\frac{c\, I^*}{u + D_I k^2}\;\nu}$$

Read it: ρ has the *opposite* sign to ν. More cells means less external resource. This is the inhibitory channel. Its magnitude is suppressed at large k — because rapid diffusion smooths out any local depletion, a short-wavelength bloom cannot dig itself a resource hole.

Step 4: Solve for σ (the byproduct response)

Linearise the R equation. Two nonlinear terms: $+cl\,In \to cl(I^*\nu + N^*\rho)$, and $-dRN \to -d(R^*\nu + N^*\sigma)$:

$$0 = cl\,(I^*\nu + N^*\rho) - r\sigma - d\,(R^*\nu + N^*\sigma) - D_R k^2 \sigma$$

Collect the σ terms, which give $-(r + dN^*)\sigma - D_R k^2 \sigma = -(v + D_R k^2)\sigma$:

$$\sigma = \frac{cl\,I^*\nu \;-\; dR^*\nu \;+\; cl\,N^*\rho}{v + D_R k^2}$$

Now handle the first two terms. Using $R^* = clKN^*/(uv)$ and $I^* = K/u$:

$$cl\,I^* - dR^* = \frac{clK}{u} - \frac{dclKN^*}{uv} = \frac{clK}{u}\left(1 - \frac{dN^*}{v}\right) = \frac{clK}{u} \cdot \frac{r}{v}$$

using $v - dN^* = r$. So

$$\boxed{\sigma = \frac{1}{v + D_R k^2}\left[\frac{clKr}{uv}\,\nu \;+\; cl\,N^*\rho\right]}$$

Read it: two contributions.

- The first is **positive**: more cells leak more byproduct. This is the activating channel.
- The second inherits ρ, which is negative. More cells deplete I, so *less* gets leaked into R. This is an inhibitory contribution travelling through the byproduct.

That second piece is the one that will matter most, so let's track it carefully.

Step 5: The N equation

Linearise. The perturbation ν multiplies the bracket — but **the bracket vanishes at the fixed point** (Step 1). So that term dies, and only N^* times the bracket's perturbation survives:

$$\lambda\nu = N^*\left[c(1-l)\rho + d\sigma\right] - D_N k^2 \nu$$

This is the crucial structural fact: N **has no autonomous linear dynamics.** Its entire growth rate is inherited from the resource perturbations. Everything downstream follows from this.

Step 6: Assemble

Substitute σ from Step 4:

$$\lambda = -D_N k^2 + \frac{N^*}{\nu}\left[c(1-l)\rho + \frac{d}{v + D_R k^2}\left(\frac{clKr}{uv}\nu + cl\,N^*\rho\right)\right]$$

Now substitute $\rho = -\dfrac{cK/u}{u + D_I k^2}\nu$ in the two places it appears, and divide out ν:

$$\lambda = -D_N k^2 \underbrace{-\frac{N^* c^2(1-l)K}{u\,(u + D_I k^2)}}_{\text{direct competition}} \underbrace{+\frac{N^* dclKr}{uv\,(v + D_R k^2)}}_{\text{cross-feeding gain}} \underbrace{-\frac{N^{*2} dc^2 lK}{u\,(u + D_I k^2)\,(v + D_R k^2)}}_{\text{indirect: depletion}\to\text{less leakage}}$$

Step 7: Compact form

Factor each denominator to expose its structure. Define the two **screening lengths**

$$\ell_I = \sqrt{\frac{D_I}{u}}, \qquad \ell_R = \sqrt{\frac{D_R}{v}}$$

so that $u + D_I k^2 = u\left(1 + \ell_I^2 k^2\right)$ and $v + D_R k^2 = v\left(1 + \ell_R^2 k^2\right)$. Then, defining

$$A = \frac{N^* dclKr}{uv^2}, \qquad I_1 = \frac{N^* c^2(1-l)K}{u^2}, \qquad I_2 = \frac{N^{*2} dc^2 lK}{u^2 v}$$

we arrive at

$$\boxed{\lambda(k) = -D_N k^2 \;+\; \frac{A}{1 + \ell_R^2 k^2} \;-\; \frac{I_1}{1 + \ell_I^2 k^2} \;-\; \frac{I_2}{\left(1 + \ell_I^2 k^2\right)\left(1 + \ell_R^2 k^2\right)}}$$

All three of A, I_1, I_2 are **positive**. This is an activator–inhibitor dispersion relation: one gain term, two loss terms.

Step 8: Sanity checks

At $k = 0$: all Lorentzians equal 1, so $\lambda(0) = A - I_1 - I_2$. This must be negative for the homogeneous state to be stable — and indeed it equals $N^* G'(N^*)$, the well-mixed eigenvalue. Net feedback is inhibitory in bulk.

At $k \to \infty$ with $D_N = 0$: every term vanishes, $\lambda \to 0$. This is Step 5's consequence: screen out the resources and N has nothing left to do.

Step 9: What drives the instability

Since $\lambda(0) < 0$ and $\lambda(\infty) = 0$ (for $D_N = 0$), instability is decided by *how* zero is approached. Expand at large k:

$$\frac{A}{1+\ell_R^2 k^2} \sim \frac{A}{\ell_R^2 k^2}, \qquad \frac{I_1}{1+\ell_I^2 k^2} \sim \frac{I_1}{\ell_I^2 k^2}, \qquad \frac{I_2}{(1+\ell_I^2 k^2)(1+\ell_R^2 k^2)} \sim \frac{I_2}{\ell_I^2 \ell_R^2 k^4}$$

The key observation: I_2 decays as k^{-4}, one power faster than the other two. It is a *double* Lorentzian, because that inhibitory signal must pass through **two** screened resource fields in sequence — depletion of I, propagated into R. In real space this means its range is $\sqrt{\ell_I^2 + \ell_R^2}$, strictly longer than the activator's ℓ_R, **for any diffusivities whatsoever**.

So the separation of scales here is **topological, not diffusive** — it comes from path length through the metabolic chain, not from D_I exceeding D_R. Dropping I_2 from the large-k balance:

$$\lambda(k) \approx \frac{1}{k^2}\left(\frac{A}{\ell_R^2} - \frac{I_1}{\ell_I^2}\right) = \frac{1}{k^2}\left(\frac{Av}{D_R} - \frac{I_1 u}{D_I}\right)$$

Instability requires this to be positive. Substituting the definitions, $Av/D_R = \dfrac{N^* dclKr}{uvD_R}$ and $I_1 u/D_I = \dfrac{N^* c^2(1-l)K}{uD_I}$, so the condition is

$$\boxed{\frac{1}{D_R}\cdot\frac{dl\,r}{v} \;>\; \frac{1}{D_I}\cdot c(1-l)}$$

For $D_I = D_R$ this reduces to the reaction-only condition $\dfrac{dlr}{r+dN^*} > c(1-l)$ — strong leakage and weak direct competition — with no diffusivity requirement, which is the case you checked against $Kc/(mr) < 1/(1-l)$.

Two necessary ingredients, distinct from each other:

1. **Sign:** the above inequality, which asks whether the net feedback flips positive once the resources are screened. Mostly a statement about reaction kinetics.
2. **Scale:** at least one $D > 0$, so that screening actually happens as k grows. If $D_I = D_R = 0$ there is no k-dependence at all and $\lambda(k) \equiv \lambda_0 < 0$ everywhere.

And D_N must be small: it is the only term that *damps* rather than merely screening, and a large enough D_N kills the band regardless of everything else. With $D_N > 0$ the unstable band is finite and a genuine wavelength is selected; with $D_N = 0$ the band extends to $k \to \infty$ and the selected scale must come from elsewhere.